

Cornell CS 3/5780

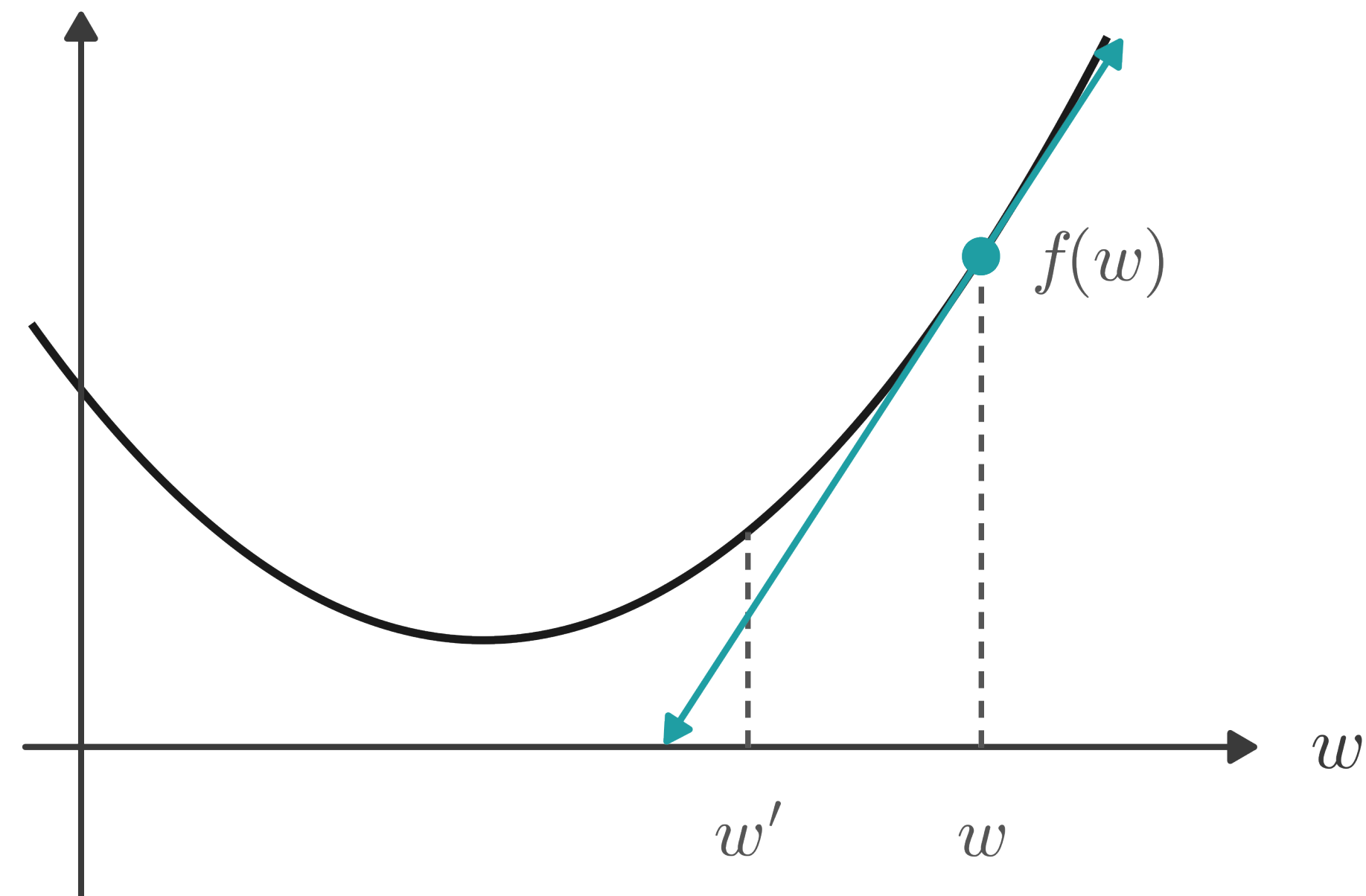
Introduction to Machine Learning

Gradient Descent II

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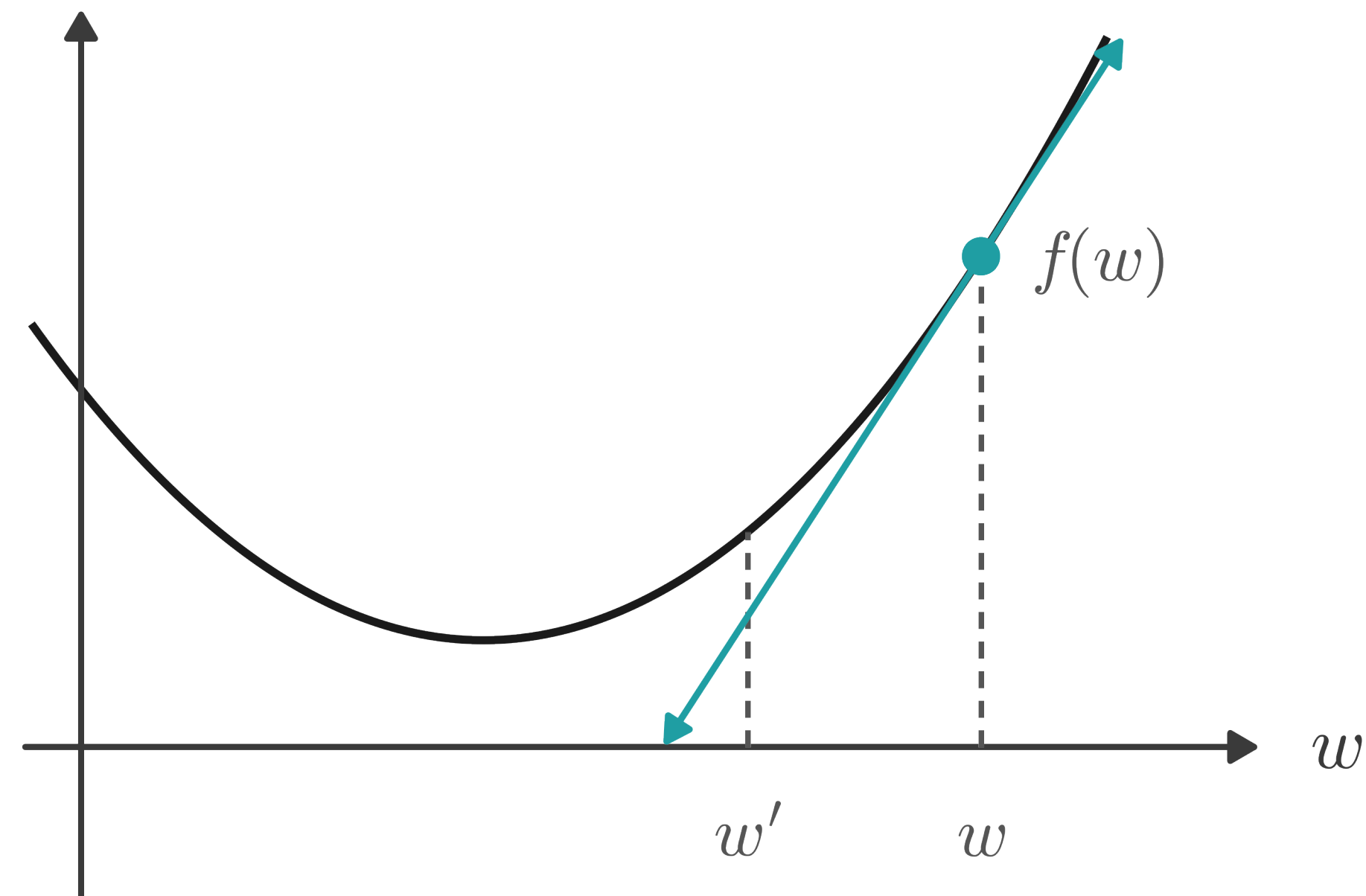
Smooth Convex Functions

- A smooth multivariate function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **convex** if and only if
$$f(w') \geq f(w) + \nabla f(w) \cdot (w' - w)$$
- For a convex function, all critical points are global minima.
 - ➔ $f(w)$ is minimized when $\nabla f(w) = 0$.



Single-Variable Smooth Convex Functions

- A smooth multivariate function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if and only if
$$f(w') \geq f(w) + f'(w)(w' - w)$$
- For a convex function, all critical points are global minima.
 - ➔ $f(w)$ is minimized when $f'(w) = 0$.



Gradient Descent

Finding the minimum of a smooth convex function

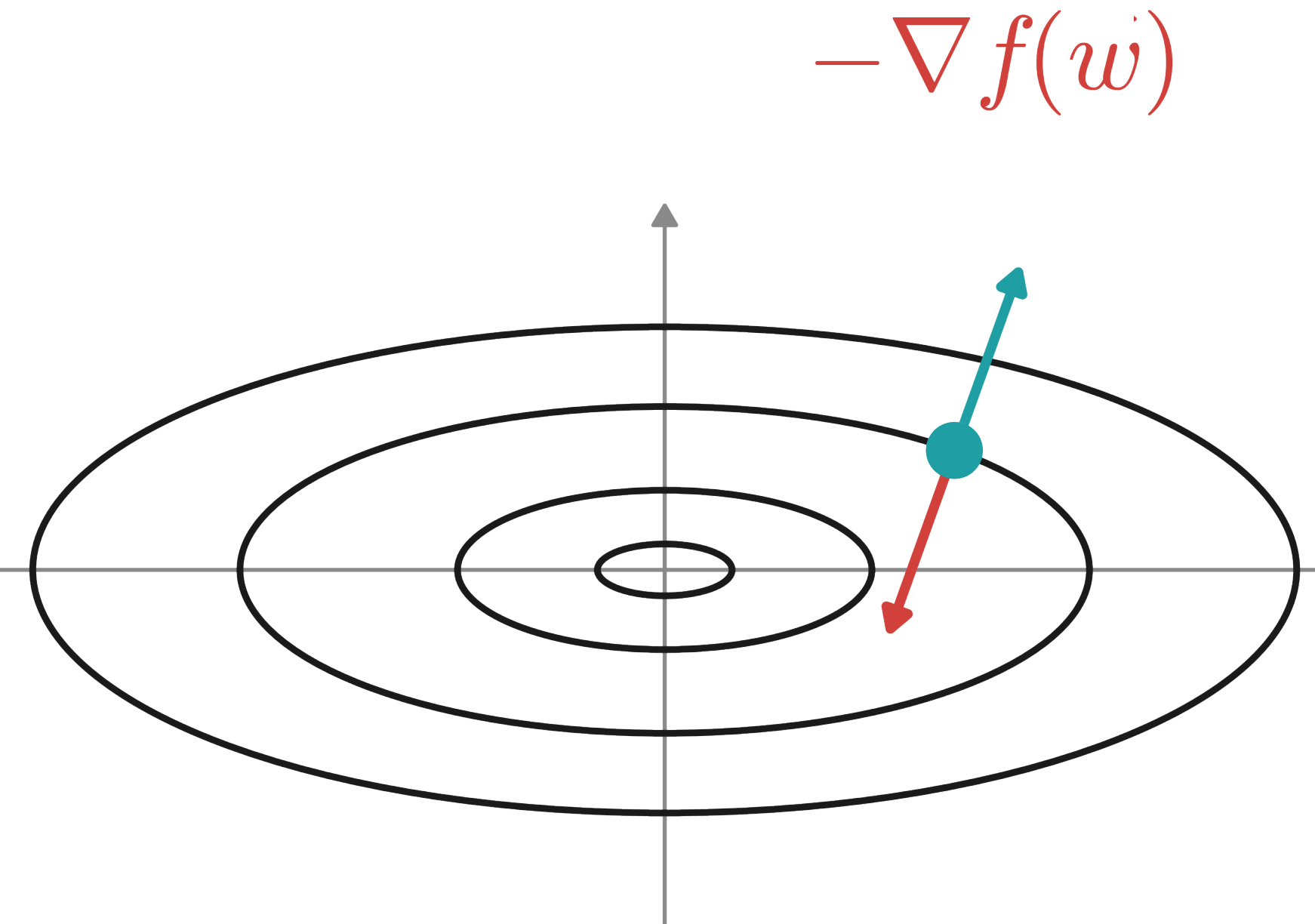
Algorithm: Gradient Descent

Input: a function f , time T , step size η_t

Initialize: $w^{(0)} = (0, \dots, 0)$

Loop: for $t = 1, 2, \dots, T$:

$$w^{(t+1)} = w^{(t)} - \eta_t \nabla f(w^{(t)})$$



Gradient Descent makes Progress

Definition: a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth if

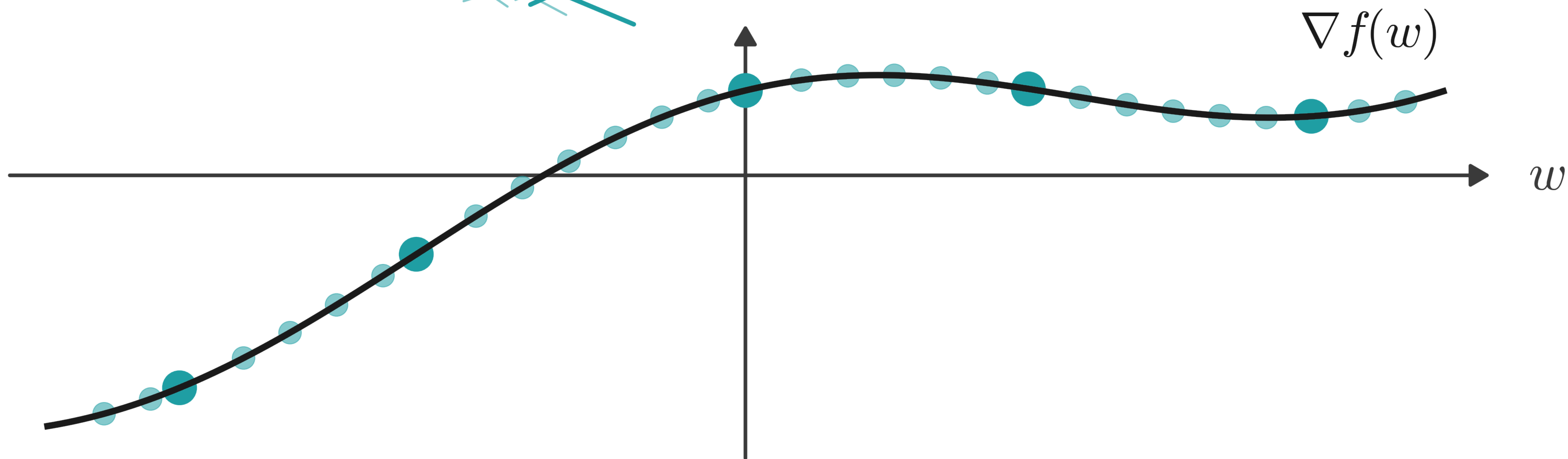
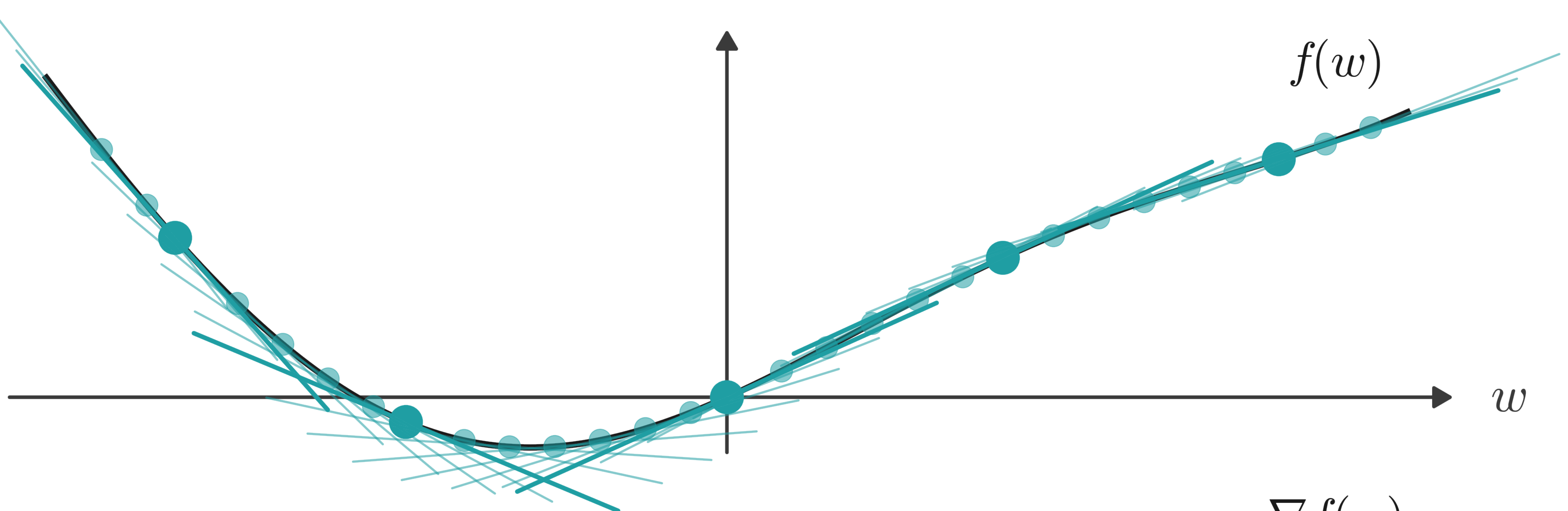
$$\|\nabla f(w) - \nabla f(w')\| \leq L\|w - w'\| \quad \text{for all } w, w'.$$

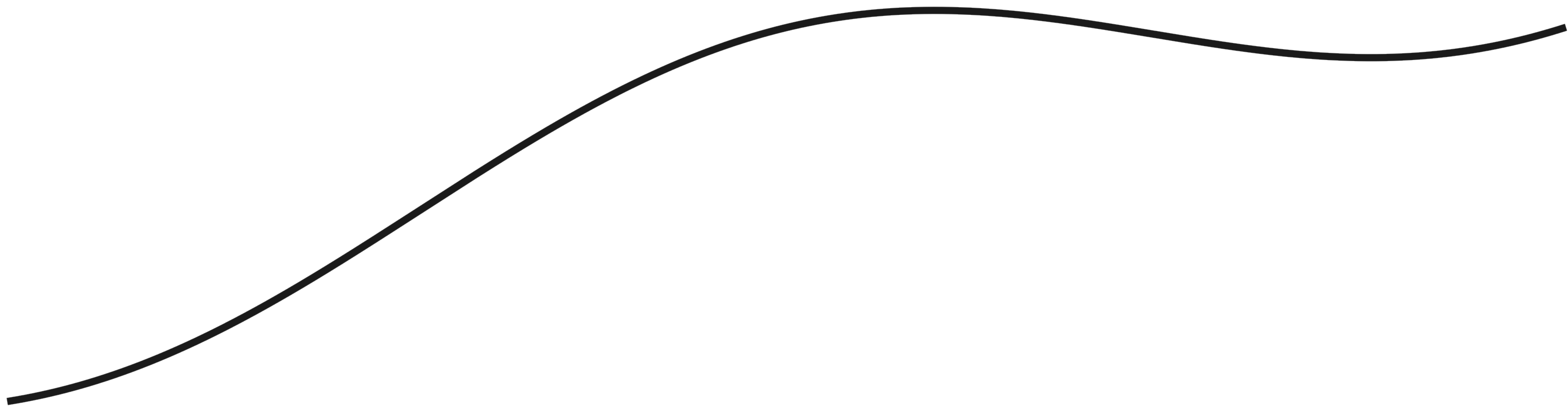
Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth.

Run gradient descent with step size $\eta_t < 1/L$.

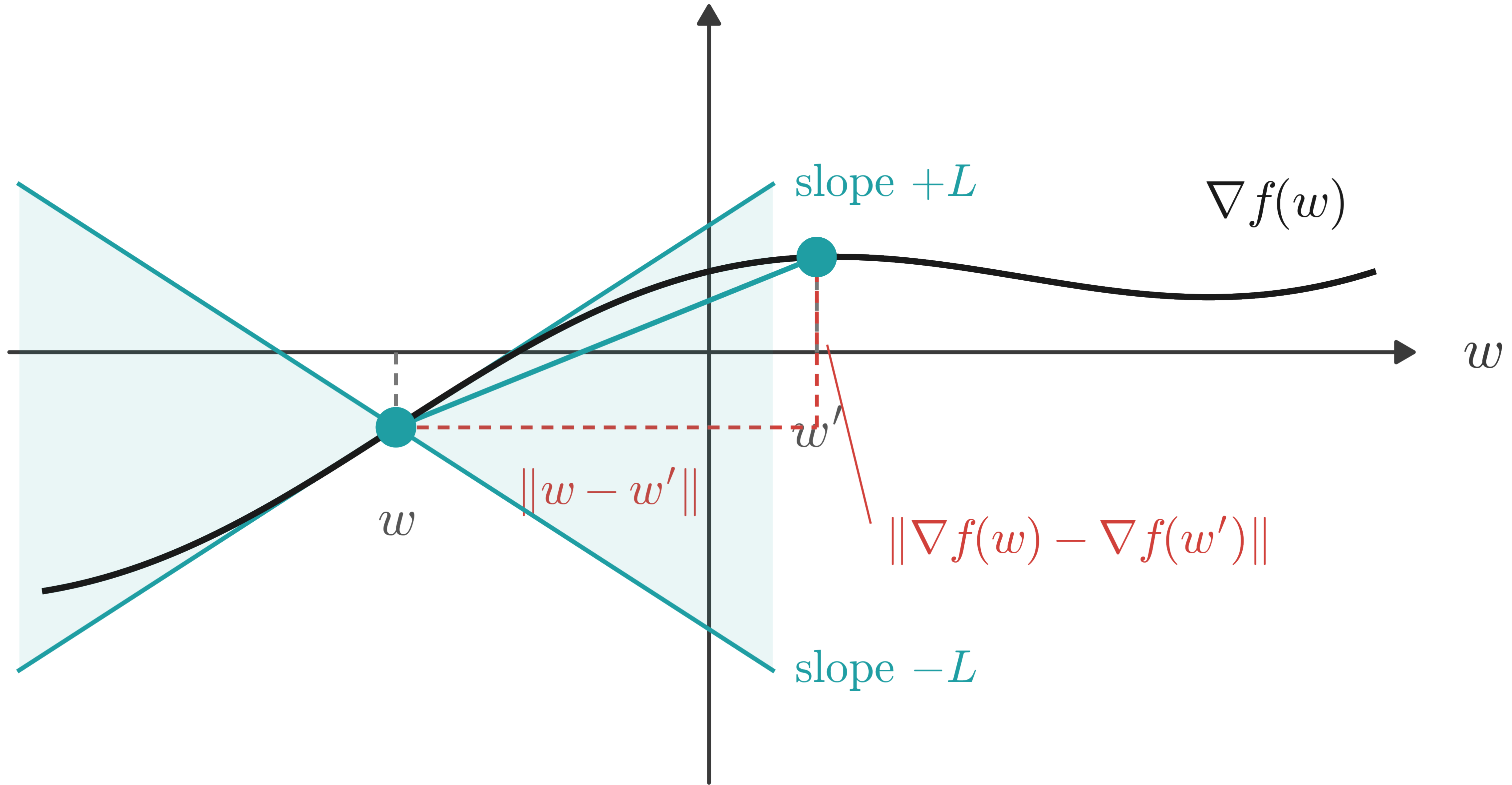
Then for every step t ,

$$f(w^{(t+1)}) \leq f(w^{(t)}) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2.$$

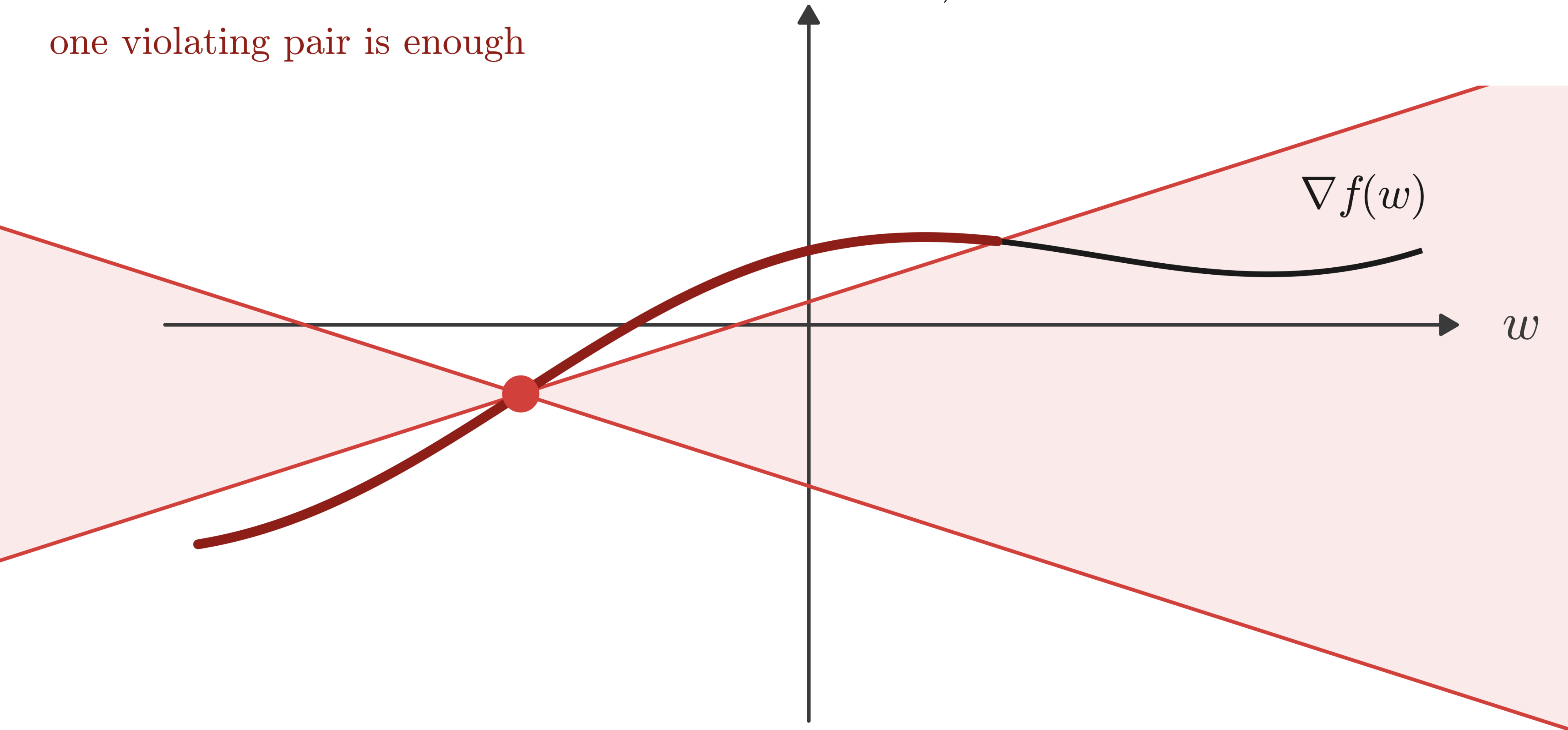




$$\|\nabla f(w) - \nabla f(w')\| \leq L \|w - w'\|$$



Not M -smooth: the bound must hold for all w, w'
one violating pair is enough



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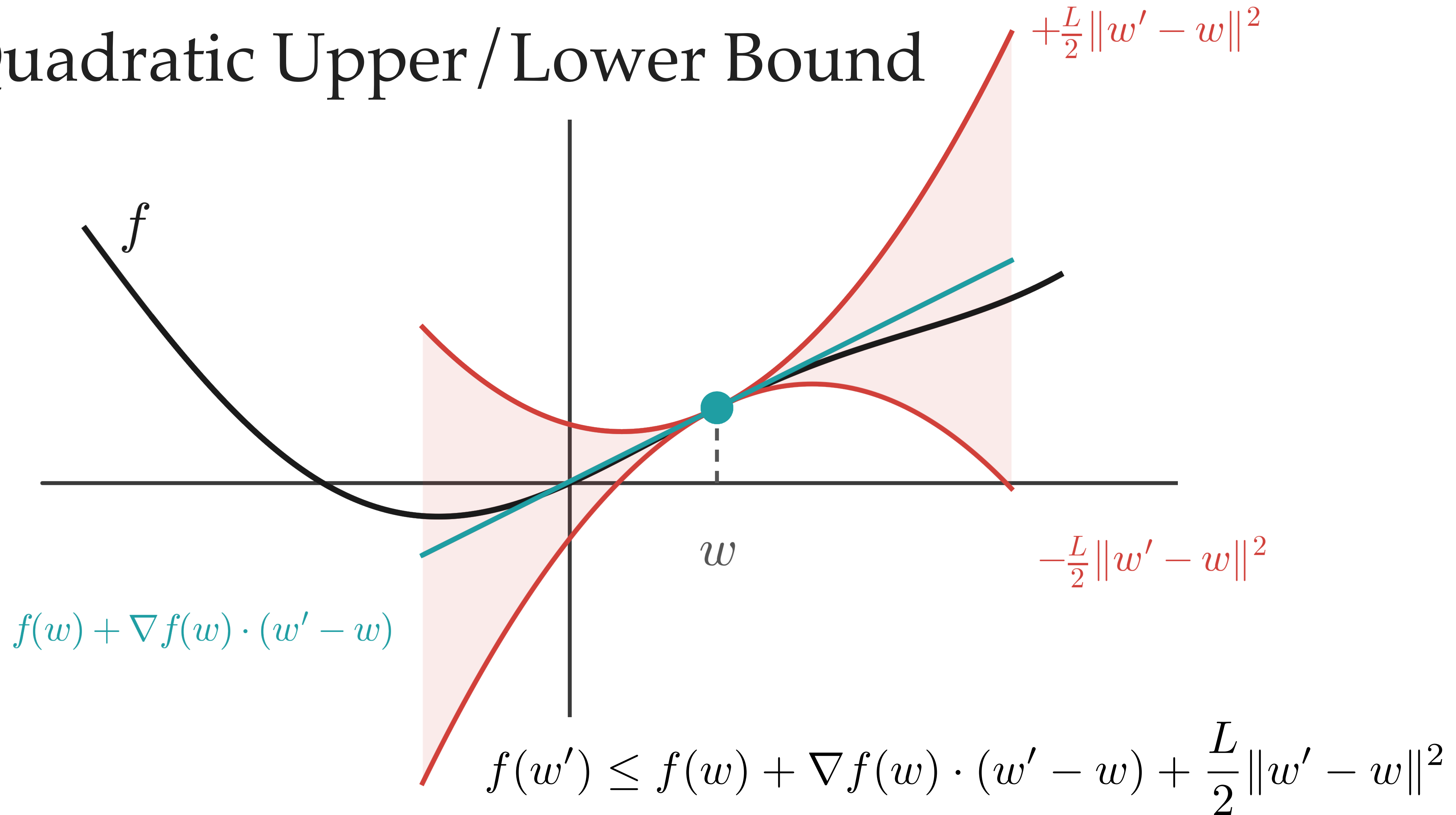
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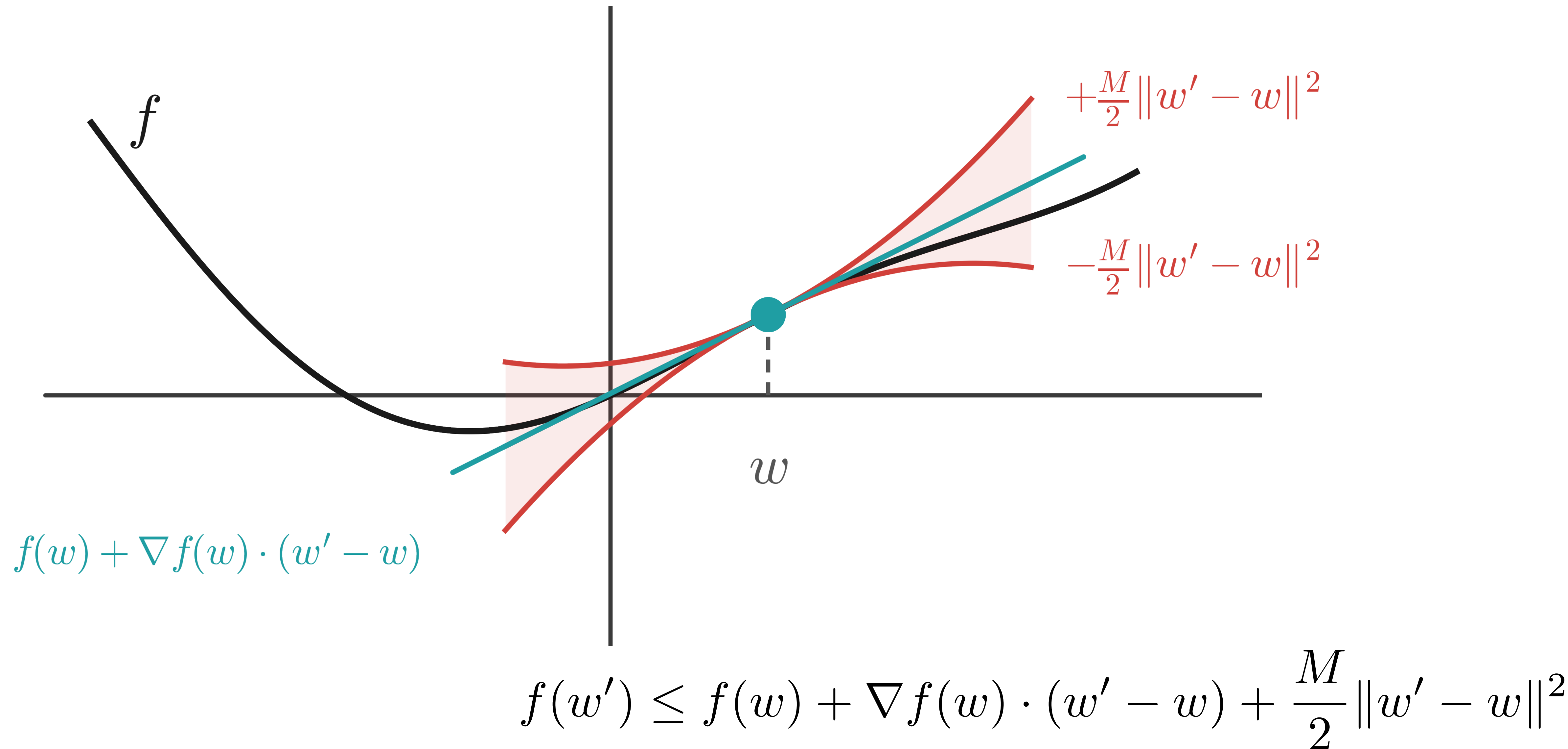
Lemma: if f is L -smooth then, for all w, w' ,

$$f(w') \leq f(w) + \nabla f(w) \cdot (w' - w) + \frac{L}{2} \|w' - w\|^2.$$

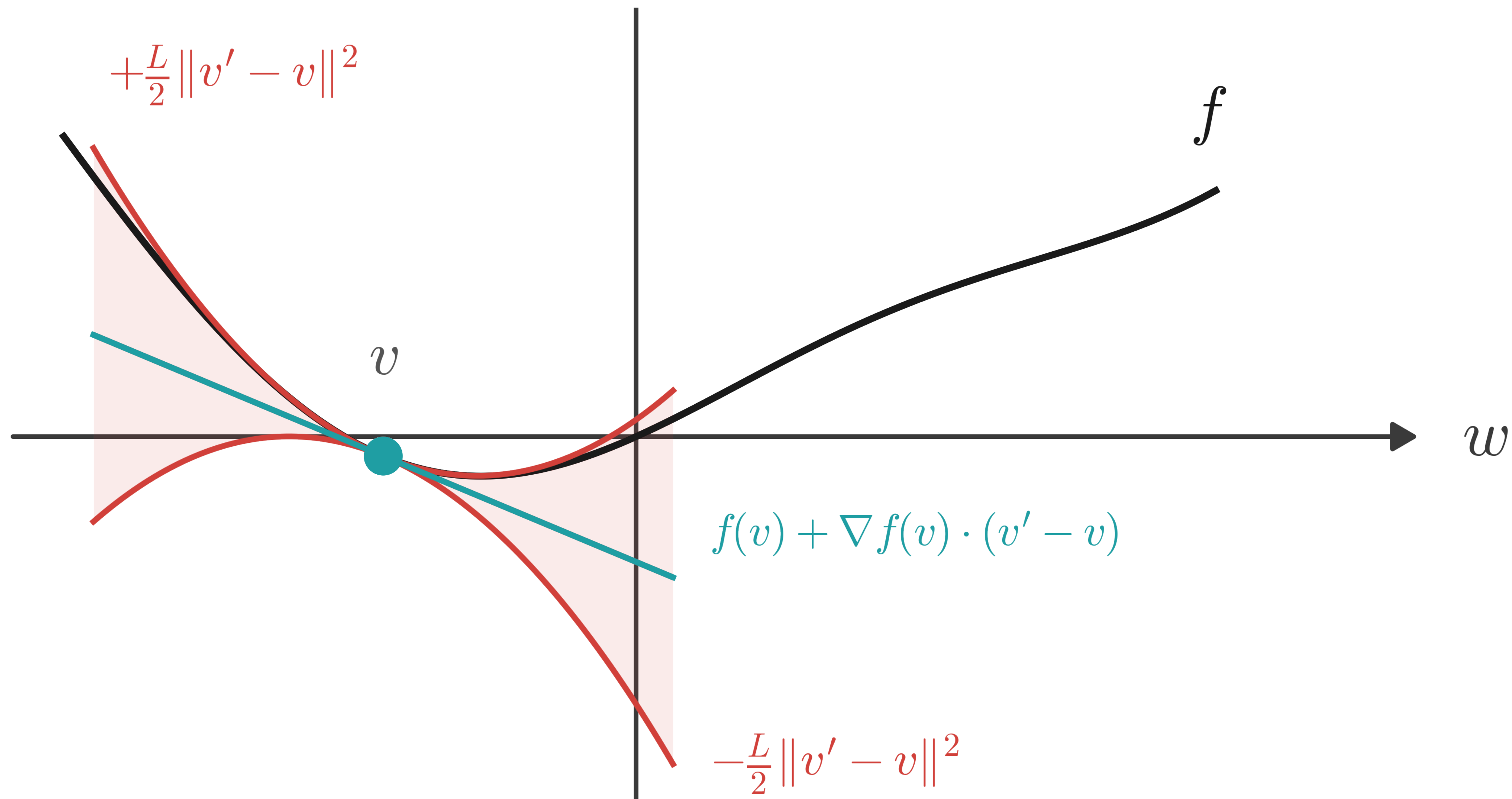
Quadratic Upper/Lower Bound



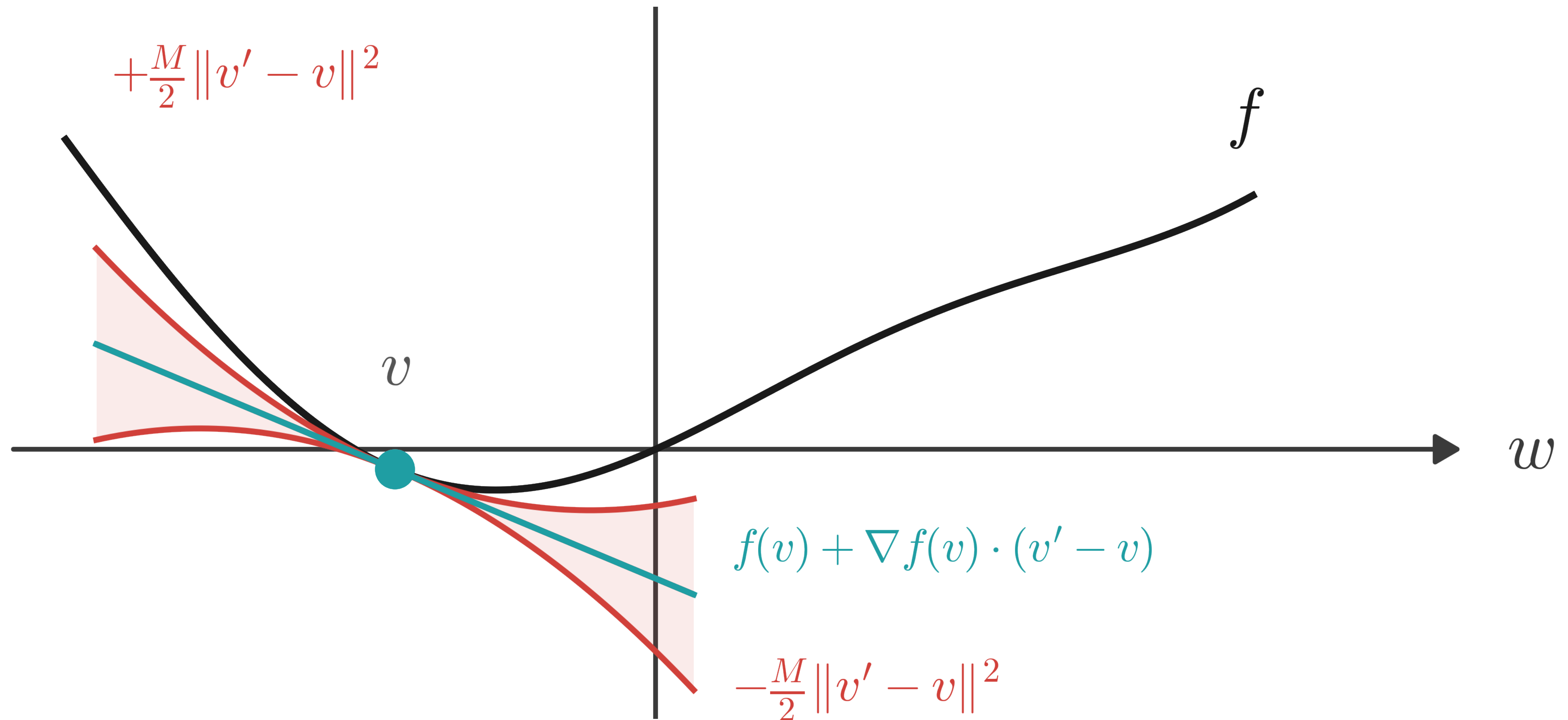
Quadratic Upper/Lower Bound



Quadratic Upper/Lower Bound



Quadratic Upper/Lower Bound



$$f(v') \not\leq f(v) + \nabla f(v) \cdot (v' - v) + \frac{M}{2} \|v' - v\|^2$$

Gradient Descent makes Progress

Definition: a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth if

$$\|\nabla f(w) - \nabla f(w')\| \leq L\|w - w'\| \quad \text{for all } w, w'.$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth.

Run gradient descent with step size $\eta_t < 1/L$.

Then for every step t ,

$$f(w^{(t+1)}) \leq f(w^{(t)}) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2.$$

Lemma: if f is L -smooth then, for all w, w' ,

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Gradient Descent makes Progress

Lemma: if f is L -smooth then, for all w, w' ,

$$f(w') \leq f(w) + \nabla f(w) \cdot (w' - w) + \frac{L}{2} \|w' - w\|^2.$$

Proof: Let $d = w' - w$ and define $g(t) = f(w + td)$ for $t \in [0, 1]$.

$$g'(t) = \nabla f(w + td) \cdot d \quad (\text{chain rule})$$

$$f(w') - f(w) = g(1) - g(0)$$

$$= \int_0^1 \nabla f(w + td) \cdot d \, dt \quad (\text{fundamental theorem of calculus})$$

$$f(w') - f(w) - \nabla f(w) \cdot d = \int_0^1 (\nabla f(w + td) - \nabla f(w)) \cdot d \, dt$$

Gradient Descent makes Progress

Lemma: if f is L -smooth then, for all w, w' ,

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$$f(w') - f(w) - \nabla f(w) \cdot d = \int_0^1 (\nabla f(w + td) - \nabla f(w)) \cdot d \, dt$$

$$\begin{aligned} (\nabla f(w + td) - \nabla f(w)) \cdot d &\leq \|\nabla f(w + td) - \nabla f(w)\| \|d\| \quad (\text{cosine} < 1) \\ &\leq L\|(w + td) - w\| \|d\| \quad (f \text{ is } L\text{-smooth}) \\ &= Lt\|d\|^2 \end{aligned}$$

Gradient Descent makes Progress

Lemma: if f is L -smooth then, for all w, w' ,

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$$f(w') - f(w) - \nabla f(w) \cdot d = \int_0^1 (\nabla f(w + td) - \nabla f(w)) \cdot d \, dt$$

$$(\nabla f(w + td) - \nabla f(w)) \cdot d \leq Lt \|d\|^2$$

$$f(w') - f(w) - \nabla f(w) \cdot d \leq \int_0^1 Lt \|d\|^2 \, dt = \frac{L}{2} \|d\|^2 = \frac{L}{2} \|w' - w\|^2$$

Gradient Descent on Smooth Convex Functions

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^\star) \leq \frac{L \|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

➡ Equivalently, suppose we want to get within ε distance of $w^\star$.

➡ I.e., we want $f(w^{(T)}) - f(w^\star) \leq \varepsilon$.

➡ We need to take no more than $T = O(1/\varepsilon)$ steps.

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^\star) \leq \frac{L \|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer w^* .

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^*) \leq \frac{L \|w^{(0)} - w^*\|^2}{2T} = O(1/T).$$

For every step t ,

$$f(w^{(t)}) \leq f(w^*) + \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) \quad (\text{convexity})$$

$$f(w^{(t+1)}) \leq f(w^{(t)}) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2 \quad (\text{smooth descent})$$

$$\leq f(w^*) + \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

$$f(w^{(t+1)}) - f(w^*) \leq \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

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$$f(w^{(t+1)}) - f(w^\star) \leq \nabla f(w^{(t)}) \cdot (w^{(t)} - w^\star) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer w^* .

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^*) \leq \frac{L \|w^{(0)} - w^*\|^2}{2T} = O(1/T).$$

$$f(w^{(t+1)}) - f(w^*) \leq \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

$$\|w^{(t+1)} - w^*\|^2 = \|w^{(t)} - \eta \nabla f(w^{(t)}) - w^*\|^2$$

$$= \left((w^{(t)} - w^*) - \eta \nabla f(w^{(t)}) \right) \cdot \left((w^{(t)} - w^*) - \eta \nabla f(w^{(t)}) \right)$$

$$= \|w^{(t)} - w^*\|^2 - 2\eta \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) + \eta^2 \|\nabla f(w^{(t)})\|^2$$

$$\|w^{(t)} - w^*\|^2 - \|w^{(t+1)} - w^*\|^2 = 2\eta \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \eta^2 \|\nabla f(w^{(t)})\|^2$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^\star) \leq \frac{L \|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

$$f(w^{(t+1)}) - f(w^\star) \leq \nabla f(w^{(t)}) \cdot (w^{(t)} - w^\star) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

$$\|w^{(t)} - w^\star\|^2 - \|w^{(t+1)} - w^\star\|^2 = 2\eta \nabla f(w^{(t)}) \cdot (w^{(t)} - w^\star) - \eta^2 \|\nabla f(w^{(t)})\|^2$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer w^* .

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$$f(w^{(T)}) - f(w^*) \leq \frac{L\|w^{(0)} - w^*\|^2}{2T} = O(1/T).$$

$$f(w^{(t+1)}) - f(w^*) \leq \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \frac{\eta}{2} \|\nabla f(w^{(t)})\|^2$$

$$\|w^{(t)} - w^*\|^2 - \|w^{(t+1)} - w^*\|^2 = 2\eta \nabla f(w^{(t)}) \cdot (w^{(t)} - w^*) - \eta^2 \|\nabla f(w^{(t)})\|^2$$

$$f(w^{(t+1)}) - f(w^*) \leq \frac{1}{2\eta} \left(\|w^{(t)} - w^*\|^2 - \|w^{(t+1)} - w^*\|^2 \right)$$

Summing up $t = 0, \dots, T-1$,
$$\sum_{t=0}^{T-1} (f(w^{(t+1)}) - f(w^*)) \leq \frac{\|w^{(0)} - w^*\|^2}{2\eta}$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

$$f(w^{(T)}) - f(w^\star) \leq \frac{L\|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

$$\text{Summing up } t = 0, \dots, T-1, \quad \sum_{t=0}^{T-1} (f(w^{(t)}) - f(w^\star)) \leq \frac{\|w^{(0)} - w^\star\|^2}{2\eta}$$

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

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Then after T steps,

$$f(w^{(T)}) - f(w^\star) \leq \frac{L\|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

$$\text{Summing up } t = 1, \dots, T, \quad \sum_{t=1}^T (f(w^{(t)}) - f(w^\star)) \leq \frac{\|w^{(0)} - w^\star\|^2}{2\eta}$$

$$f(w^{(T)}) - f(w^\star) \leq f(w^{(t)}) - f(w^\star) \quad (\text{smooth descent})$$

$$T \left(f(w^{(T)}) - f(w^\star) \right) \leq \sum_{t=1}^T (f(w^{(t)}) - f(w^\star))$$

$$f(w^{(T)}) - f(w^\star) \leq \frac{\|w^{(0)} - w^\star\|^2}{2T\eta} = \frac{L\|w^{(0)} - w^\star\|^2}{2T}$$

Accelerated Gradient Descent

Can we go faster than $O(1/T)$?

Algorithm: Nesterov Momentum

Input: a function f , time T , step size η , β_t

Initialize: $w^{(0)} = (0, \dots, 0)$

Loop: for $t = 1, 2, \dots, T$:

$$y^{(t)} = w^{(t)} + \beta_t (w^{(t)} - w^{(t-1)})$$

$$w^{(t+1)} = y^{(t)} - \eta \nabla f(y^{(t)})$$

➡ This is an example of “momentum” methods

➡ Bias your next step in the direction of the previous step $w^{(t)} - w^{(t-1)}$

➡ Amount of momentum is controlled by β_t

Comparing Rates

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth and **convex** with minimizer $w^\star$.

Run gradient descent with step size $\eta_t = 1/L$.

Then after T steps,

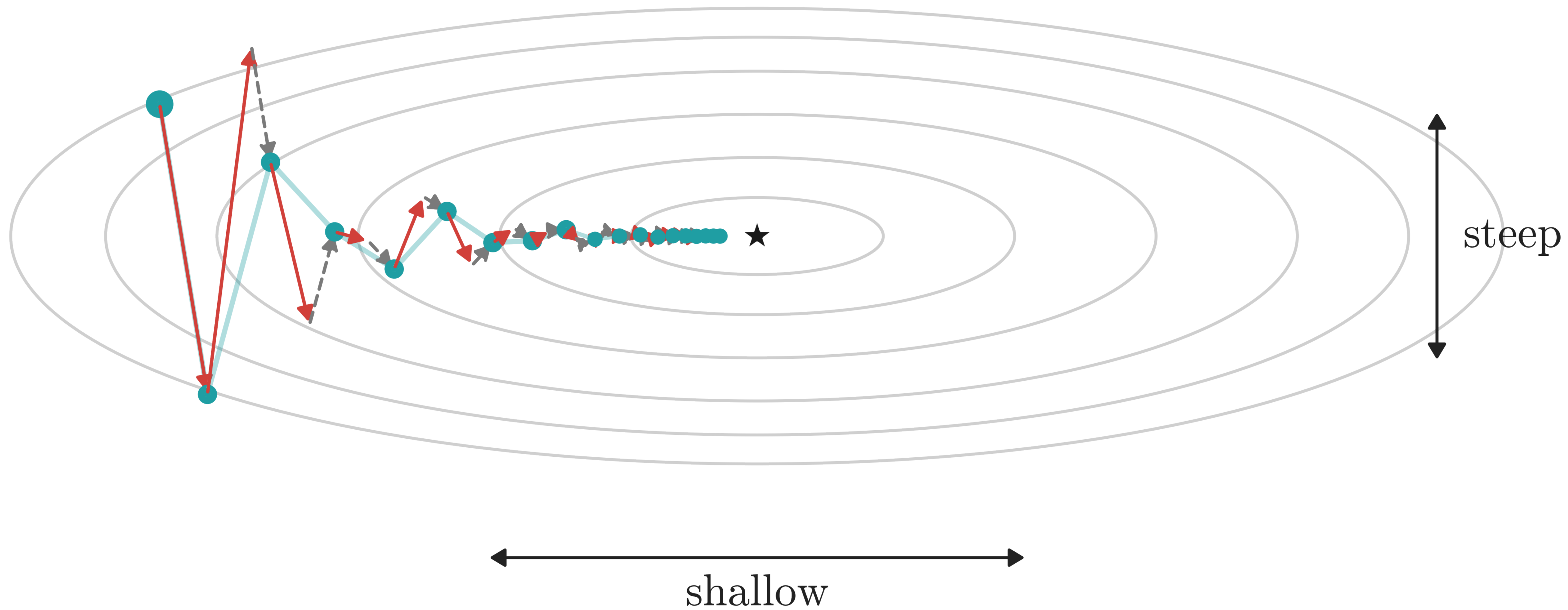
$$f(w^{(T)}) - f(w^\star) \leq \frac{L \|w^{(0)} - w^\star\|^2}{2T} = O(1/T).$$

Running Nesterov accelerated gradient step with $\eta_t = 1/L$, $\beta_t = \frac{t-1}{t+2}$,

$$f(w^{(T)}) - f(w^\star) \leq \frac{2L \|w^{(0)} - w^\star\|^2}{(T+1)^2} = O(1/T^2).$$

➡ We need to take no more than $T = O(1/\sqrt{\varepsilon})$ steps.

— gradient step - - - momentum — actual step



Adaptive Gradients + Momentum

Algorithm: Adam

Input: a function f , time T , step size η , momentum β_1, β_2 , and ε

Initialize: $w^{(0)} = m^{(0)} = v^{(0)} = 0$

Loop: for $t = 1, 2, \dots, T$:

$$m^{(t+1)} = \beta_1 m^{(t)} + (1 - \beta_1) \nabla f(w^{(t)}), \quad \hat{m}^{(t+1)} = \frac{m^{(t+1)}}{1 - \beta_1^{t+1}},$$

$$v^{(t+1)} = \beta_2 v^{(t)} + (1 - \beta_2) (\nabla f(w^{(t)}))^2, \quad \hat{v}^{(t+1)} = \frac{v^{(t+1)}}{1 - \beta_2^{t+1}},$$

$$w^{(t+1)} = w^{(t)} - \eta \frac{\hat{m}^{(t+1)}}{\sqrt{\hat{v}^{(t+1)} + \varepsilon}}.$$